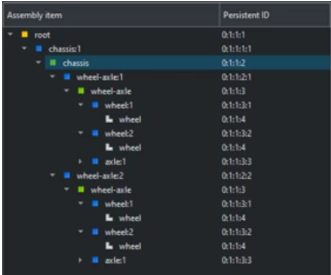
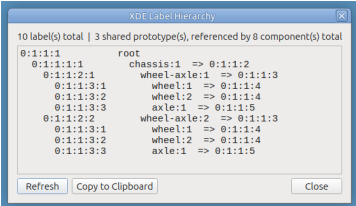
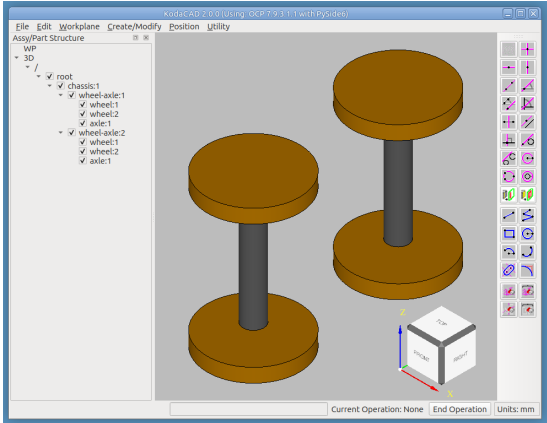


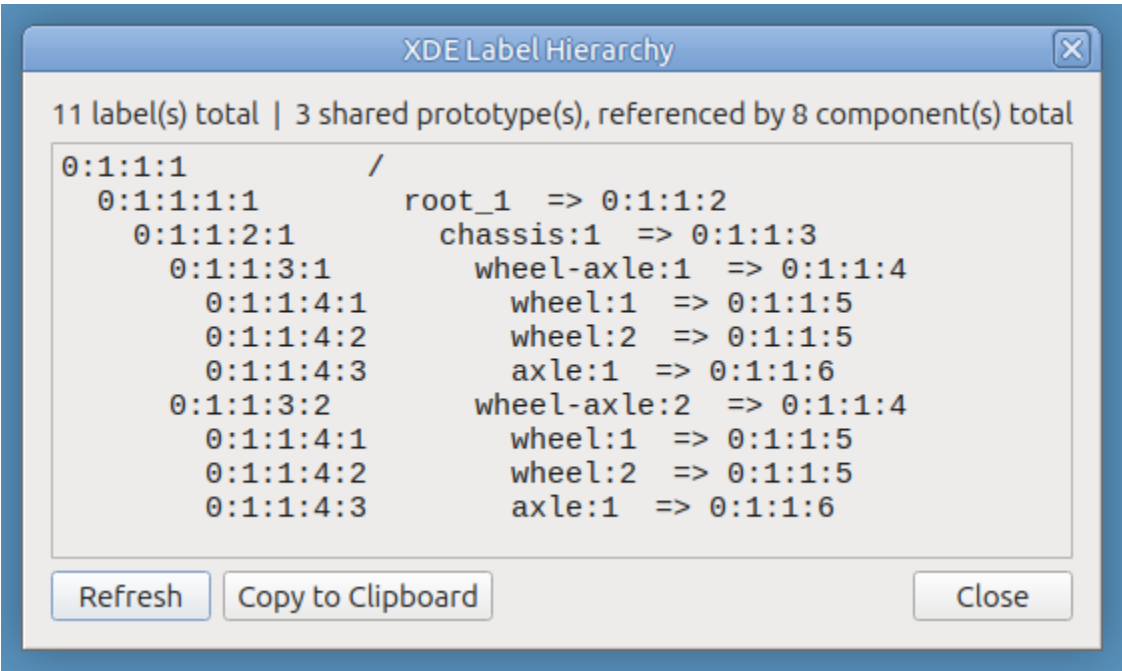
Tutorial: Building Quaoar's Chassis

This tutorial is based on Lesson 12 of the Quaoar Workshop Tutorials (https://www.youtube.com/playlist?list=PL_WFkJrQIY2iVVchOPhI77xl432jeNYfQ) series, in which a simple chassis example (chassis.step) is used to explore how parts, components, instances, prototypes, and subassemblies are represented in a hierarchical assembly in Open Cascade.

KodaCAD is built on the Open Cascade kernel, so it makes sense to benchmark its own ability to construct this chassis exactly. If chassis.step is loaded into KodaCAD as a session, the resulting assembly hierarchy corresponds exactly to the one shown in Quaoar's Lesson 12 video.

Quaoar Hierarchical Assembly	Hierarchy (in KodaCAD)	Chassis session displayed in KodaCAD
		

Alternatively, if the file is imported into an empty KodaCAD session, the root label is brought in as a **child** of / , as shown below.



What you'll need

The `chassis.step` file in this folder, for Step 1. Steps 2 onward build a wheel-and-axle assembly entirely from scratch in a fresh KodaCAD session -- no file needed for those.

What this exercises

Measuring 3D geometry from an imported STEP file with the calculator's Ctrl+Shift circle-center picking; native part creation via sketching and Pull; sibling-assembly creation; drag-reparenting parts into an assembly; shared-instance creation; most of the Position dialog's surface (Dynamic, Nudge, Mate Align, 2 Points) for positioning both a shared part and a shared assembly; STEP save; and Undo/Redo across a long, mixed chain of native-construction, assembly-structure, and positioning operations.

Step 1 -- Measure dimensions of the 3D model in the original file `chassis.step`

1. Load `chassis.step` as a session, or import it into an existing one.
2. Use the calculator's bottom-row buttons to measure the 3D model.
 - Pick the center of a circle by holding down **Ctrl+Shift**.

Here are the dimensions measured this way:

Part	Radius (mm)	Axial Length (mm)
Axle	7.5	110
Wheel	33.288	10

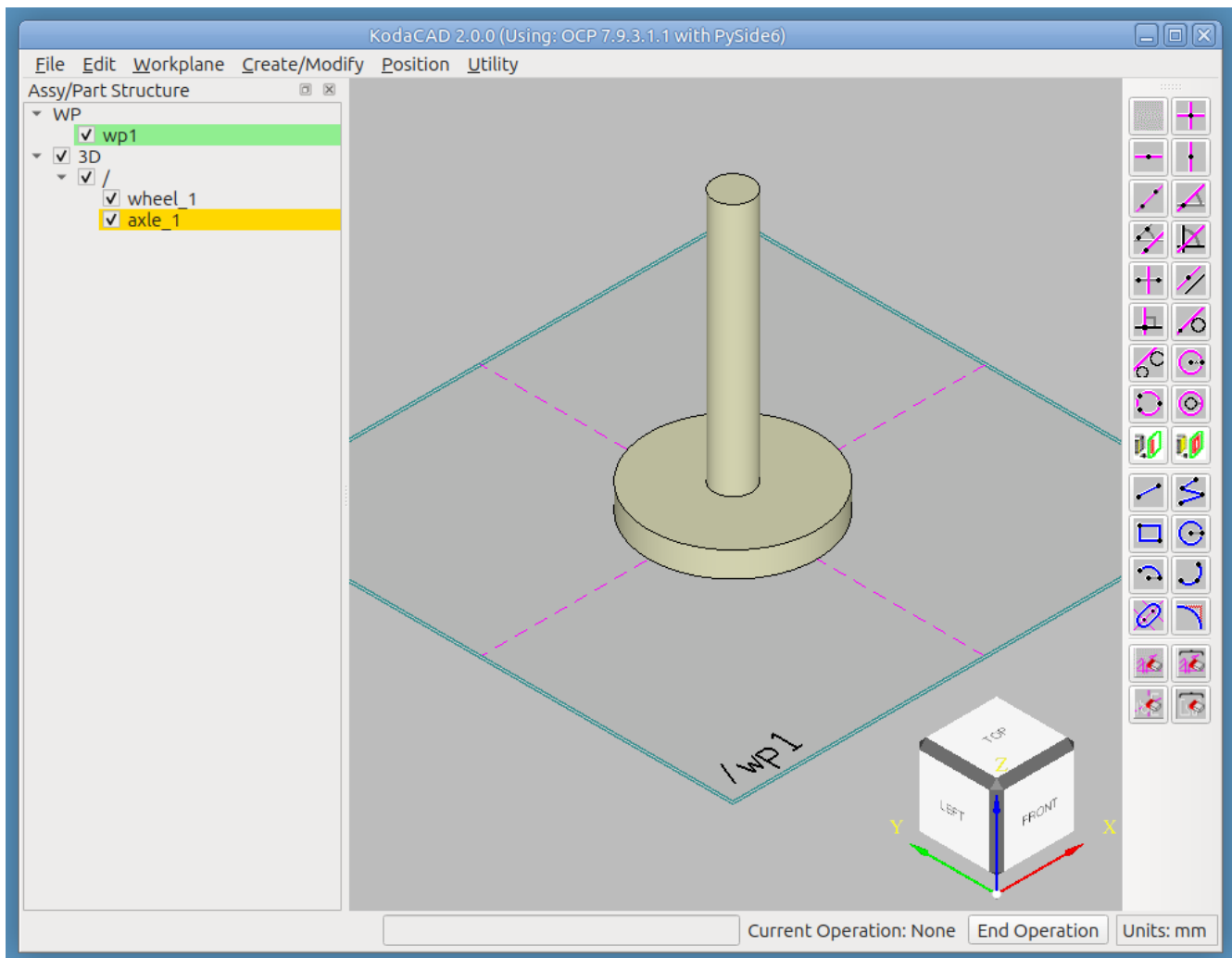
Distance between axles = 105 mm

Exercises: STEP loading (either path); the calculator's Ctrl+Shift circle-center picking against real, imported 3D geometry.

Step 2 -- Create a wheel and an axle

- Start KodaCAD.
- Create a new workplane, **At Origin** on the XY Plane.
- Click the top face of the ViewCube for a straight-down view filling the viewport.
- Use the **Circle** tool to draw a circle at the origin with the axle radius.
- Draw a second, concentric circle with the wheel radius.
- RMB click on '/' and select **Create Empty Part**
 - Name the part 'wheel'
 - **Create/Modify -> Pull**, Complete the dialog:

- Operation: Add Material
 - Mode: Linear
 - Direction: +W
 - Distance 10
- The wheel is created (A hole for the axle is a deliberate design choice here, not required.)
- Temporarily hide the wheel so only the workplane is displayed.
 - Use the **Delete Geometry Element** tool to delete the outer circle.
 - Now create an axle using the same process used to create the wheel
 - Clicking the ViewCube's top-left-front corner gives this view:

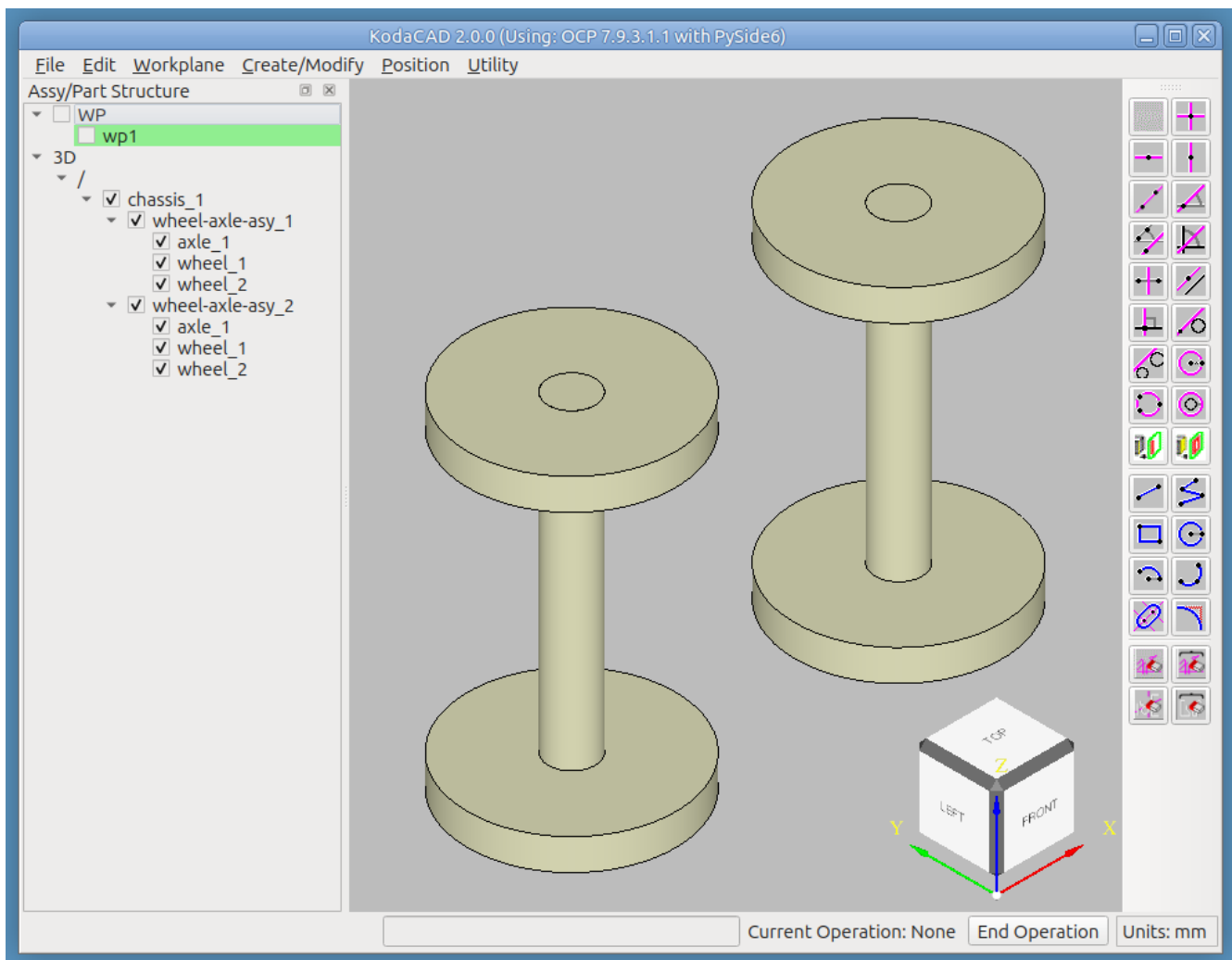


Exercises: workplane creation, concentric circle sketching, hiding a part mid-workflow, Delete Geometry Element, and two independent Pull operations from the same sketch.

Step 3 -- Create the assembly structure

- RMB click / in the tree and select **Create New Assembly**. Name it "wheel-axle-asy".
- Drag "axle" into "wheel-axle-asy" in the tree.

- Drag "wheel" into "wheel-axle-asy" in the tree.
- RMB click "wheel" and select **Create Shared Instance**.
 - Rename it to "wheel_2" via RMB.
- Select the new wheel instance and position it at the opposite end of the axle. All three methods work:
 - **Dynamic** -> Nudge 100 mm in Z.
 - **Mate Align** -> Align. Click top face of wheel, then top face of axle.
 - **2 Points** -- Pick center of wheel top, then center of axle top.
- RMB click / and select **Create New Assembly** again. Name it "chassis".
- Drag "wheel-axle-asy" into "chassis".
- RMB click "wheel-axle-asy" and select **Create Shared Instance**.
 - Rename it to "wheel-axle-asy_2".
- Select the new assembly instance and position it 105 mm in the X direction. (In the original file, the second assembly is moved in Y instead – either works to demonstrate the mechanism.)



Exercises: creating a new assembly directly at the root / in a fresh session; drag-reparenting parts into an assembly; shared-instance creation for both a part and an assembly; all three Position dialog methods

(Dynamic/Nudge, Mate Align, 2 Points), including 2-Points picking a circle's center on a non-analytic (heavily-cylindrical, likely NurbsConverted) 3D edge.

Step 4 -- Save the session

KodaCAD doesn't have its own native save format -- it depends entirely on saving and loading a session as a STEP file.

Exercises: STEP save on a from-scratch, multi-level assembly with two shared instances.

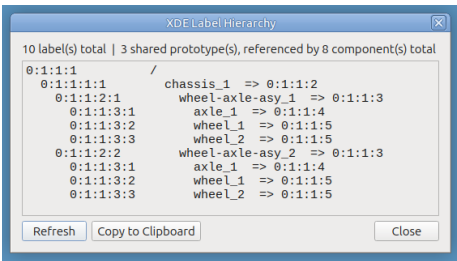
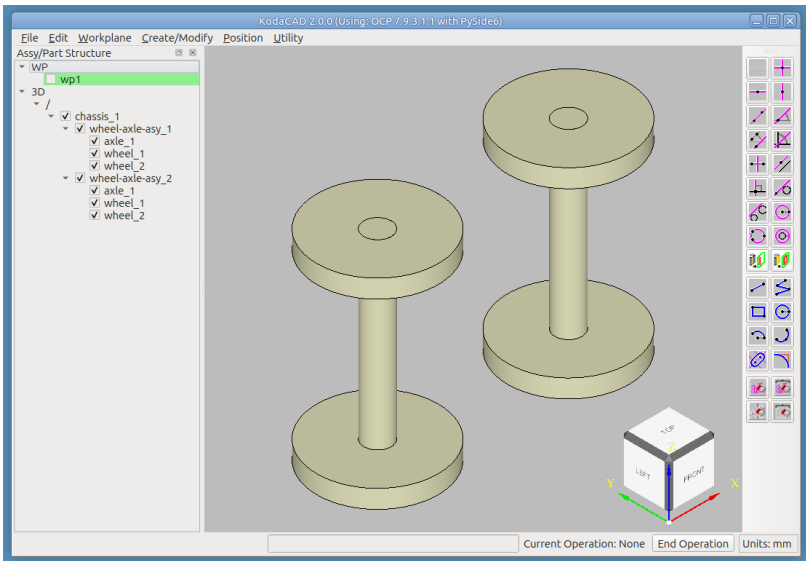
Step 5 -- Test Undo / Redo

- Before exiting this session, check how many Undo steps are available: **Utility -> Undo/Redo count**. Expect somewhere around 14.
- Use Undo several times and watch the model step back through earlier states.
- Confirm Redo returns the session to the completed state, one step at a time.

Exercises: undo/redo across a long, mixed chain -- native part creation, assembly-structure changes (which have to create the root / label the very first time), reparenting, shared-instance creation, and positioning, all in the same undo history.

Step 6 -- Reload the saved session

- Undo/Redo history doesn't survive exiting a session.
- Workplane(s) are lost too -- they're pure UI state, never written to the OCAF document at all.
- The reloaded 3D model itself is restored 100%.

Hierarchy (in KodaCAD)	Chassis displayed in KodaCAD
	

Exercises: STEP reload fidelity for a from-scratch, multi-level assembly with shared instances -- confirming

the round trip preserves everything Step 3 built.

Notes for whoever runs this next

- If any step's status bar message, menu label, or tool name has changed since this was written, that's worth fixing here directly – this document drifting out of sync with the actual UI is exactly the kind of thing it's meant to catch.
- See the Assembly Structure tutorial ([../as1-oc-214/README.md](#)) for the two STEP-loading paths explored in more depth, and for fillet propagation across a *pre-existing* (rather than freshly-created) shared instance.
- See the Lathe tutorial ([../lathe/README.md](#)) for organizing and navigating a large, messily-imported assembly – renaming, sibling assemblies, and reparenting an assembly that already has a parent to a different one.